

EXP NO.8 Consider a library Management System. Develop a javaScript program that will validate the controls in the forms you have created for the application. State the assumptions you make (business logic you are taking into consideration)

Aim

To develop a Library Management System that validates form inputs using JavaScript and stores/retrieves data from a database using Servlet/JSP technology.

Procedure

1. Create a Dynamic Web Project and configure Apache Tomcat.
2. Create a MySQL database and required table (books).
3. Design a JSP form to collect library details.
4. Implement JavaScript validation for form inputs.
5. Create a Servlet to handle form submission.
6. Retrieve form data using `request.getParameter()`.
7. Connect to the database using JDBC.
8. Insert validated data into the database using `PreparedStatement`.
9. Deploy and run the application on the server.
10. Test validation and verify data storage in the database.

Code:

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<title>Add Book</title>
```

```
<style>
```

```
body{font-family:Arial;background:#f4f6f9;text-align:center;}
```

```
form{background:white;width:300px;margin:auto;padding:20px;border-radius:10px;}
```

```
input{margin:10px;padding:8px;width:90%;}

button{padding:10px;background:#4CAF50;color:white;border:none;}

</style>

</head>

<body>

<h2>Add Book</h2>

<form action="addbook" method="post">

<input type="text" name="title" placeholder="Title" required>

<input type="text" name="author" placeholder="Author" required>

<input type="number" name="price" placeholder="Price" required>

<button type="submit">Add</button>

</form>

</body>

</html>
```

```
package com.library.servlet;

import java.io.*;

import jakarta.servlet.*;

import jakarta.servlet.http.*;

import jakarta.servlet.annotation.WebServlet;

import java.sql.*;
```

```
@WebServlet("/addbook")

public class AddBookServlet extends HttpServlet {

    protected void doPost(HttpServletRequest request, HttpServletResponse response)

        throws ServletException, IOException {

        String title = request.getParameter("title");

        String author = request.getParameter("author");

        String price = request.getParameter("price");


        try{

            Class.forName("com.mysql.cj.jdbc.Driver");


            Connection con = DriverManager.getConnection(

                "jdbc:mysql://localhost:3306/library_db",

                "root",

                "YOUR_PASSWORD");


            PreparedStatement ps = con.prepareStatement(

                "INSERT INTO books(title,author,price) VALUES(?,?,?)");

            ps.setString(1,title);

            ps.setString(2,author);

            ps.setDouble(3,Double.parseDouble(price));

            ps.executeUpdate();


            response.sendRedirect("books");

        }catch(Exception e){
```

```
e.printStackTrace();

}

}

}

<!DOCTYPE html>

<html>

<head>

<title>Library System</title>

<style>

body{font-family:Arial;background:#f4f6f9;text-align:center;}

button{padding:12px 25px;background:#4CAF50;color:white;border:none;border-radius:5px;}

</style>

</head>

<body>

<h1>Library Management System</h1>

<a href="books"><button>View Books</button></a>

<br><br>

<a href="addbook.html"><button>Add Book</button></a>

</body>

</html>

package com.library.servlet;

import java.io.*;

import jakarta.servlet.*;

import jakarta.servlet.http.*;
```

```
import jakarta.servlet.annotation.WebServlet;

import java.sql.*;

@WebServlet("/books")

public class ViewBooksServlet extends HttpServlet {

    protected void doGet(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {

        response.setContentType("text/html");

        PrintWriter out = response.getWriter();

        out.println("<html><body>");

        out.println("<h2>Library Books</h2>");

        try{

            Class.forName("com.mysql.cj.jdbc.Driver");

            Connection con = DriverManager.getConnection(

                "jdbc:mysql://localhost:3306/library_db",

                "root",

                "YOUR_PASSWORD");

            Statement st = con.createStatement();

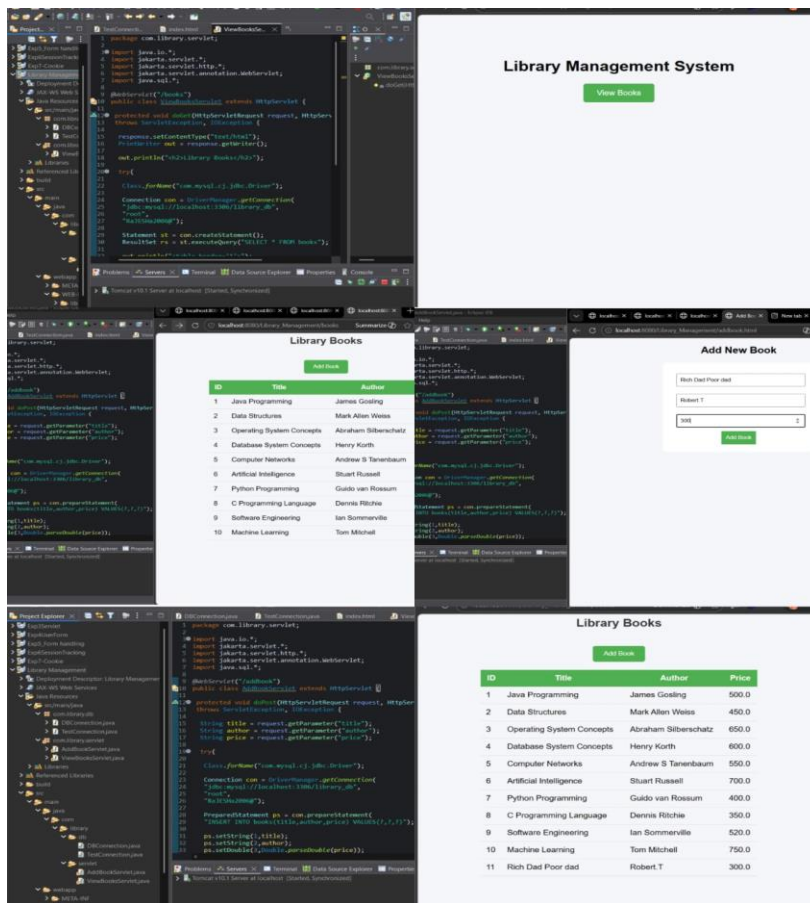
            ResultSet rs = st.executeQuery("SELECT * FROM books");

            out.println("<table border='1'>");

            out.println("<tr><th>ID</th><th>Title</th><th>Author</th><th>Price</th></tr>");
```

```
while(rs.next()){  
    out.println("<tr>");  
    out.println("<td>"+rs.getInt("id")+"</td>");  
    out.println("<td>"+rs.getString("title")+"</td>");  
    out.println("<td>"+rs.getString("author")+"</td>");  
    out.println("<td>"+rs.getDouble("price")+"</td>");  
    out.println("</tr>");  
}  
out.println("</table>");  
}catch(Exception e){  
    e.printStackTrace();  
}  
out.println("</body></html>");  
}  
}
```

Output:



Result

The Library Management System was successfully developed. JavaScript validation ensured correct user input before submission, and the Servlet/JSP application successfully stored and retrieved data from the database.